We know that the perpendicular distance from the given point with position vector $\vec{u}$ to the plane $\vec{r}\cdot\vec{n}=p$ is given by $\delta=\dfrac{|\vec{u}\cdot\vec{n}-p|}{|\vec{n}|}$. Therefore, substituting $\vec{u}=(2,5,-3)=2\hat{i}+5\hat{j}-3\hat{k}$ and $\vec{n}=6\hat{i}-3\hat{j}+2\hat{k}$ in the formula, we get

$$\delta=\frac{|\vec{u}\cdot\vec{n}-p|}{|\vec{n}|}=\frac{\left|\left(2\hat{i}+5\hat{j}-3\hat{k}\right)\cdot\left(6\hat{i}-3\hat{j}+2\hat{k}\right)-5\right|}{\left|6\hat{i}-3\hat{j}+2\hat{k}\right|}=2 \text{ units.}$$

Example 6.50

Find the distance of the point $(5,-5,-10)$ from the point of intersection of a straight line passing through the points $A(4,1,2)$ and $B(7,5,4)$ with the plane $x-y+z=5$.

Solution

The Cartesian equation of the straight line joining A and B is

$$\frac{x-4}{3}=\frac{y-1}{4}=\frac{z-2}{2}=t \quad \text{(say)}.$$

Therefore, an arbitrary point on the straight line is of the form $(3t+4,4t+1,2t+2)$. To find the point of intersection of the straight line and the plane, we substitute $x=3t+4, y=4t+1, z=2t+2$ in $x-y+z=5$, and we get $t=0$. Therefore,the point of intersection of the straight line is (4, 1, 2). Now, the distance between the two points (4, 1, 2) and $(5,-5,-10)$ is

$$\sqrt{(4-5)^2+(1+5)^2+(2+10)^2}=\sqrt{181} \text{ units.}$$

6.8.13 Distance between two parallel planes

Theorem 6.21

The distance between two parallel planes $ax+by+cz+d_1=0$ and $ax+by+cz+d_2=0$ is given by $\dfrac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$.

Proof

Let $A(x_1,y_1,z_1)$ be any point on the plane $ax+by+cz+d_2=0$, then we have

$ax_1+by_1+cz_1+d_2=0 \Rightarrow\ ax_1+by_1+cz_1=-d_2$

The distance of the plane $ax+by+cz+d_1=0$ from the point $A(x_1,y_1,z_1)$ is given by

$$\delta=\frac{|ax_1+by_1+cz_1+d_1|}{\sqrt{a^2+b^2+c^2}}=\frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$$

Hence, the distance between two parallel planes $ax+by+cz+d_1=0$ and $ax+by+cz+d_2=0$ is given by $\delta=\dfrac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$.

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